STANDARD VIII — CHEMISTRY (ANSWER KEY)

FIRST TERM SUMMATIVE ASSESSMENT 2026 | EM

Subject: Chemistry | **Class: Standard 8** | **Time: 40 Minutes** | **Total Score: 20**

<table>
<tr><th>Q. No.</th><th>Value Points / Expected Answers (Answer Only)</th><th>Score</th></tr>
<tr><td>1</td><td>(d) Rusting of iron is a chemical change. (d)</td><td>1</td></tr>
<tr><td>2</td><td>B. 1-b, 2-c, 3-d, 4-a B
(1-Metals & Non-metals, 2-Triads, 3-Law of Octaves, 4-Atomic mass)</td><td>1</td></tr>
<tr><td>3</td><td>(a) The cup becomes hot / Heat is felt.
(b) Heat is liberated during the chemical reaction between quicklime and water (Exothermic reaction).
$CaO + H_2O \rightarrow Ca(OH)_2 + Heat$</td><td>2
(1+1)</td></tr>
<tr><td>4 (A)</td><td>(a) A white precipitate of Silver Chloride is formed; it turns black/grey when exposed to sunlight.
(b) Silver chloride decomposes into metallic silver by absorbing light energy.</td><td>2
(1+1)</td></tr>
<tr><td>4 (B)</td><td>OR 4 (B)
(a) Carbon dioxide + Water ($CO_2 + H_2O$)
(b) Photochemical reaction</td><td>2
(1+1)</td></tr>
<tr><td>5</td><td>(a) Chemical energy → Electrical energy
(b) Electrical energy → Chemical energy</td><td>2
(1+1)</td></tr>
<tr><td>6</td><td>• a. All matter is made up of extremely tiny particles called atoms.
• c. Atoms of different elements differ in size, properties, and mass.</td><td>2</td></tr>
<tr><td>7 (A)</td><td>
<table>
<tr><th>Element</th><th>Basis of Naming</th><th>Symbol</th></tr>
<tr><td>Indium</td><td>(a) Colour (Indigo)</td><td>In</td></tr>
<tr><td>(b) Rutherfordium</td><td>Rutherford (Scientist)</td><td>Rf</td></tr>
<tr><td>Europium</td><td>(c) Continent (Europe)</td><td>Eu</td></tr>
<tr><td>(d) Neptunium</td><td>Planet (Neptune)</td><td>Np</td></tr>
</table>
</td><td>2</td></tr>
<tr><td>7 (B)</td><td>OR 7 (B)
<table>
<tr><th>Element</th><th>Symbol</th></tr>
<tr><td>Aluminium</td><td>Al</td></tr>
<tr><td>Sulphur</td><td>S</td></tr>
<tr><td>Gold</td><td>Au</td></tr>
<tr><td>Potassium</td><td>K</td></tr>
</table>
</td><td>2</td></tr>
<tr><td>8</td><td>2N : Represents 2 separate (individual) nitrogen atoms.
N_2 : Represents 1 molecule of nitrogen containing 2 bonded nitrogen atoms.</td><td>2</td></tr>
<tr><td>9</td><td>(a) Negative terminal (Cathode)
(b) Silver cyanide + Sodium cyanide solution (or potassium argentocyanide solution)</td><td>3
(1+1+1)</td></tr>
</table>

Q. No.	Value Points / Expected Answers (Answer Only)	Score
	(c) Yes. It is an electrochemical reaction because electrical energy is absorbed/consumed to bring about the chemical change.	
10 (A)	• **Number of molecules** = 3 • Number of Carbon atoms = $3 \times 2 = 6$ • Number of Hydrogen atoms = $3 \times 6 = 18$ • **Total number of atoms** = $6 + 18 = 24$	**3**
10 (B)	**OR 10 (B)** **a) Reactant :** $CaCO_3$ (Calcium carbonate) **b) Products :** CaO (Calcium oxide), CO_2 (Carbon dioxide) **c) Chemical equation :** $CaCO_3 \rightarrow CaO + CO_2$ (on heating)	**3** (1+1+1)

Prepared by Ravi Sir